



CSE623 Machine Learning

Weekly Report 6

Feature-Based Global Wheat Full Semantic Segmentation using Classical Machine Learning and Contour Analysis

Submitted to faculty: Prof. Mehul Raval

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Aim:

To conduct exploratory data analysis (EDA), extracting multi-dimensional feature sets (7, 16, and 25 features), and evaluating the effectiveness of classical machine learning models, especially Random Forest (RF) and Support Vector Machine (SVM).

Introduction:

We are using classical machine learning models: Random Forest and SVM, as they provide more understandable and computationally effective alternative. We use Handcrafted Feature Engineering to bridge gap, as these models cannot perceive spatial patterns. We expand feature space from 7 features to 25 features set to overcome environmental obstacles.

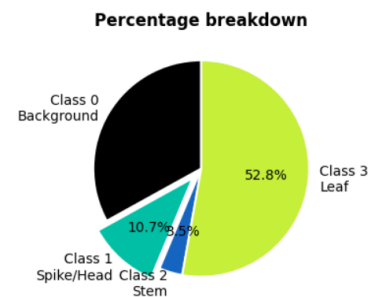
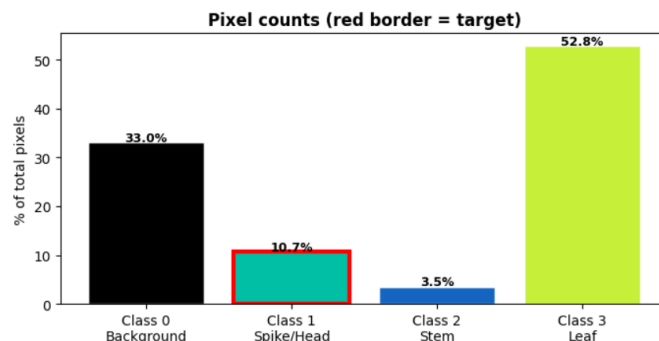
Work Completed:

EDA:

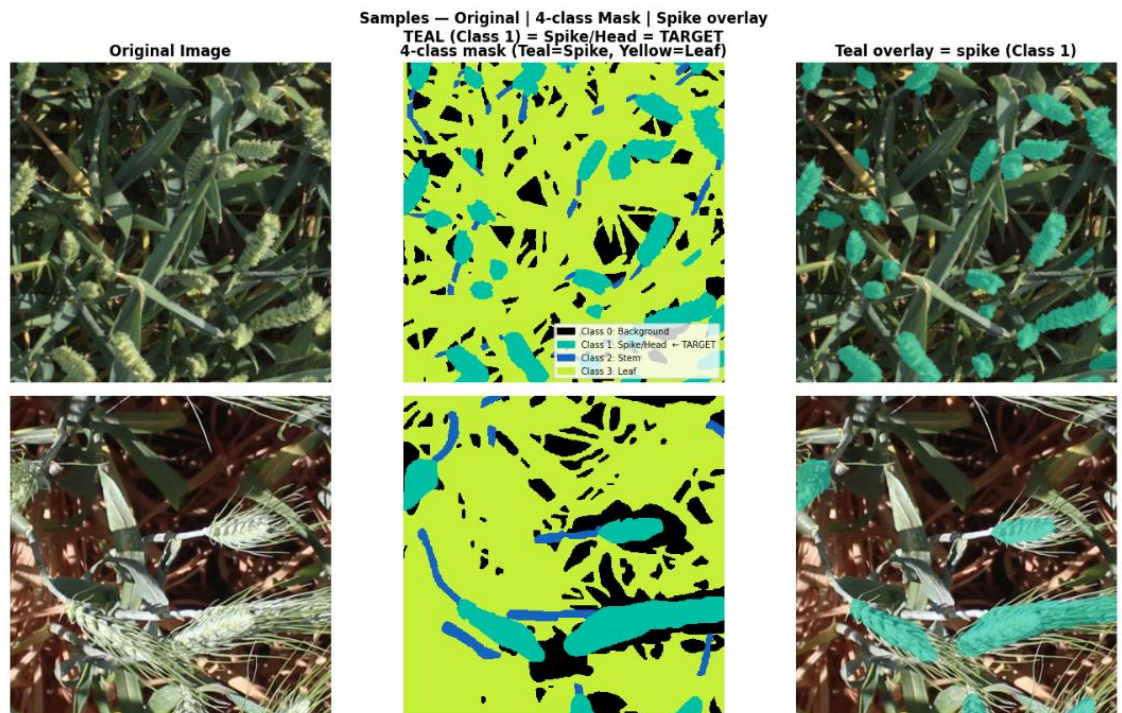
- i. **Class Distribution:** We analyzed the distribution of pixels across different classes of dataset. The majority of pixels were found to be dominated by the “Leaf” class, with the wheat heads(target) occupying a smaller portion. This shows that the dataset is imbalanced, which is usually common in segmentation tasks.

```
Pixel distribution across 99 training images:  
Class 0 Background : 33.03% #####  
Class 1 Spike/Head : 10.67% ##### <- TARGET  
Class 2 Stem      : 3.49% #  
Class 3 Leaf      : 52.82% #####
```

GWFFS — Class Pixel Distribution (99 images)



- ii. **Sample Visualization:** We analyzed sample images along with their masks and spikes overlay to understand the data better. The spike regions show correct matching with the masks, ensuring proper labeling.



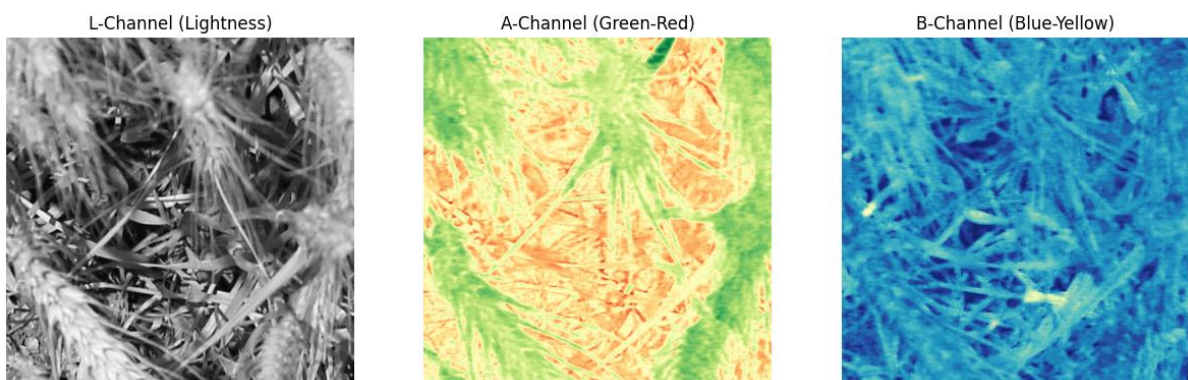
Feature Engineering:

This week, we build the Feature Engineering Pipeline, which is the core module for extracting the descriptors needed for our Classical Machine Learning Model.

Color and Chromaticity Features (9 Features):

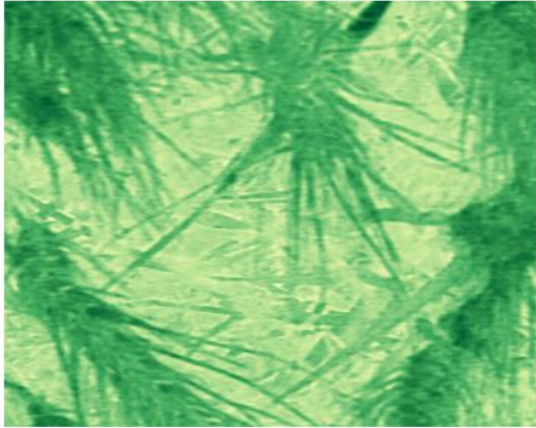
The goal was to separate plant matter from soil and use mathematics to neutralize the effects of direct sunlight and shadows.

1. **LAB Color Space (L, A, B):** It prevents shadows from obscuring the plant's actual color by separating pure Lightness (L) from the Green-Red (A) and Blue-Yellow (B) color channels.

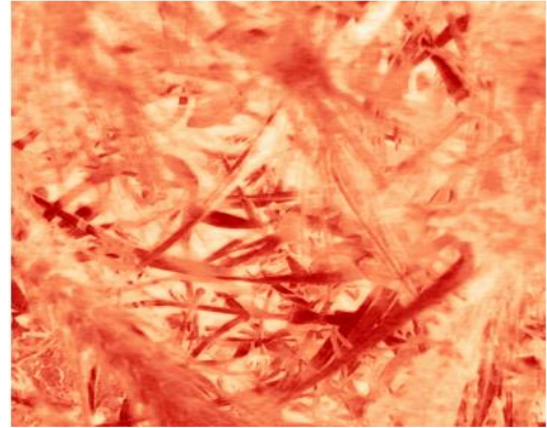


2. **Vegetation Indices (ExG, ExR):** The biological plant pixels are significantly enhanced by Excess Green and Excess Red, while background and dirt pixels are mathematically reduced.

Excess Green (ExG) - Plant Matter

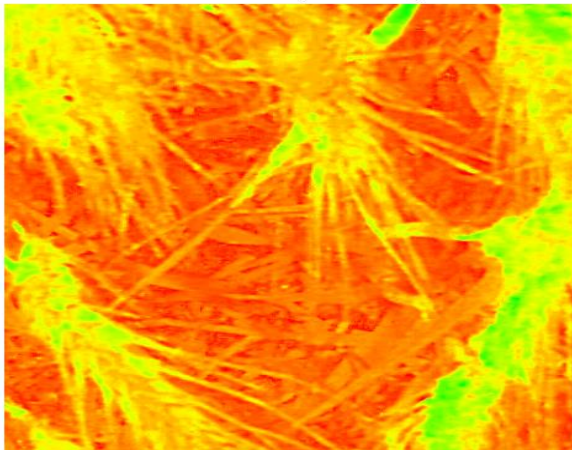


Excess Red (ExR) - Mature Heads

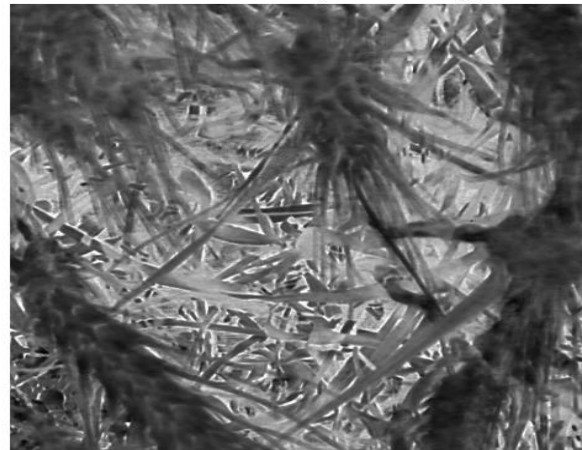


3. Shadow Invariance (H, S, Norm-R, Norm-G): In order to prevent dark wheat heads in the shadows from being mistakenly identified as soil, Hue, Saturation, and Normalized Red and Normalized Green solely depend on the pixel's base pigment (chromaticity) rather than its brightness.

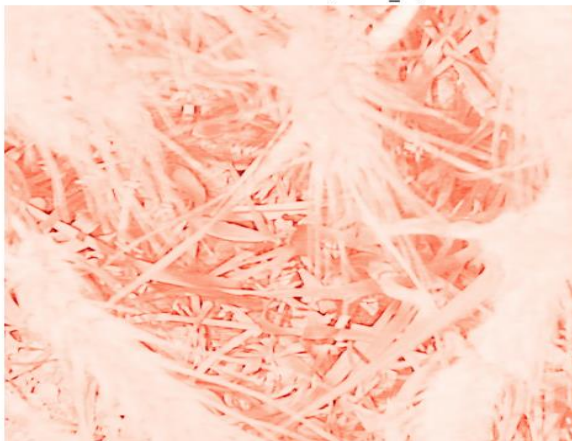
Hue Channel (Base Color)



Saturation Channel (Color Intensity)



Normalized Red (norm_R)



Normalized Green (norm_G)

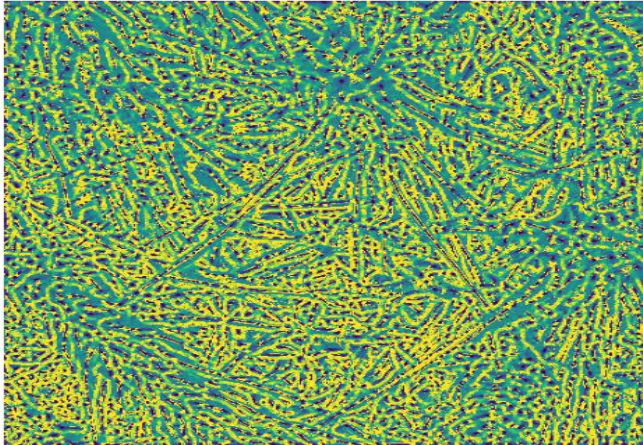


Texture and Frequency Features (9 Features):

The goal was to differentiate between the smooth surfaces of leaves and rough surfaces of the wheat spikes.

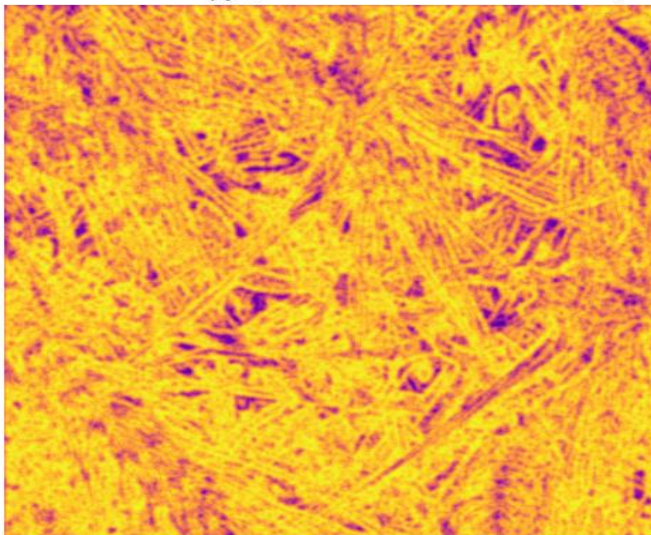
1. Local Binary patterns (LBP): This was used to capture micro-textures by comparing the pixel's intensity with that of its immediate neighbors.

LBP (Micro-texture Complexity)

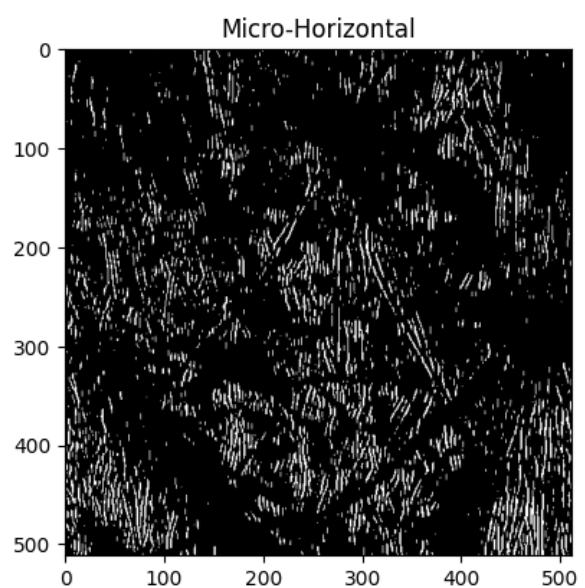
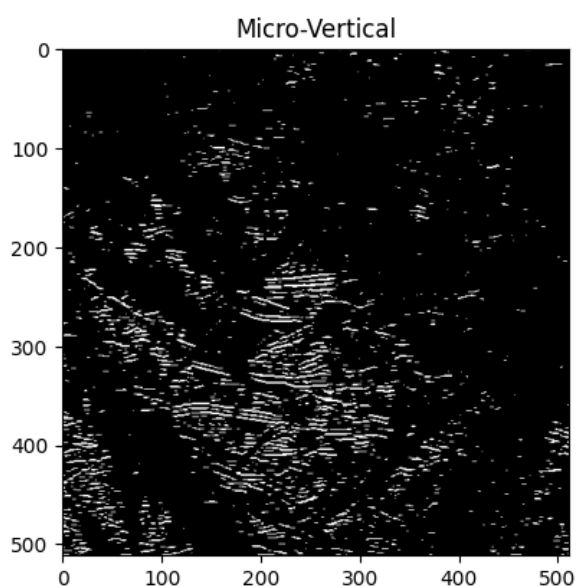
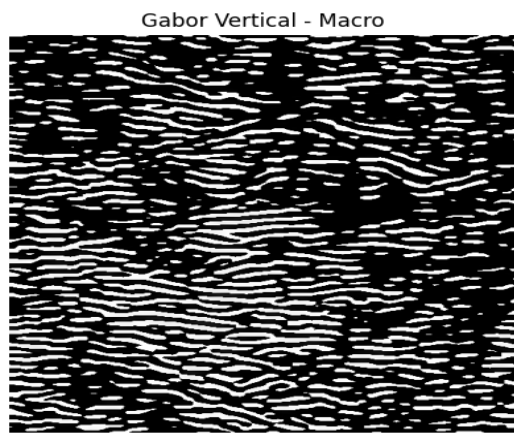


2. Shannon Entropy: This measures how complex or mathematically complex a region is. For example, wheat heads have high entropy, while flat leaves have low entropy.

Shannon Entropy (Chaotic Wheat Head Detection)



3. Gabor Filters (Macro Vertical & Horizontal, Micro Vertical & Horizontal): Here we use 4 different filters that work in different directions. “Macro” frequencies pick up broad vertical stems and horizontal leaves, while “Micro” frequencies pick up the fine, crisscrossing textures of the wheat hairs (awns).



Spatial and Statistical Context (3 Features):

To overcome the “single-pixel blindness” of classical ML, each pixel is required to be informed about its 9*9 neighborhood mathematically.

1. Gaussian Context: It is a weighted, blurred average that provides a smooth representation of the general area and reduces camera static.
2. Local Mean: It is a hard average of the grid around it that tells the model about the general brightness of the pixel’s immediate neighborhood.
3. Local Variance: It is a powerful metric measuring the roughness of the neighborhood. Smooth leaves have zero-variance, while the variance of bumpy wheat head is extremely large.

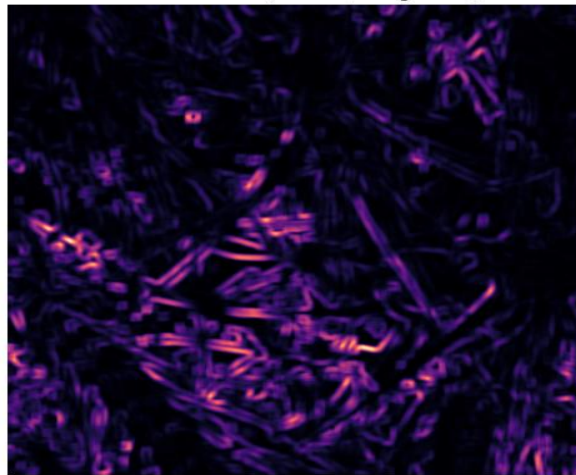
Gaussian Filter (Weighted Spatial Context)



Local Mean (Neighborhood Average)



Local Variance (Texture/Roughness)

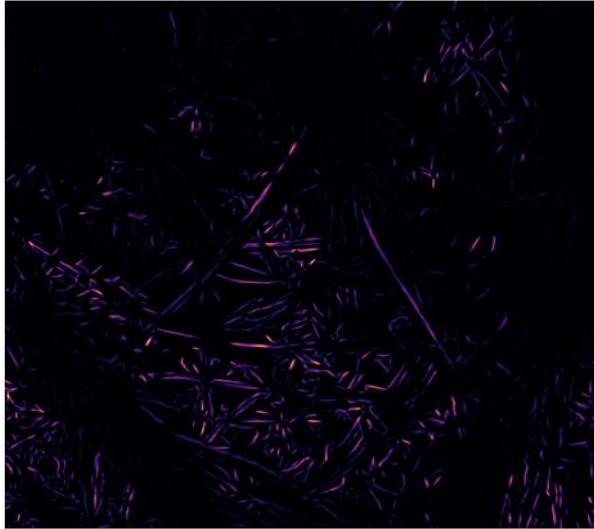


Structural and Biological Features (4 Features):

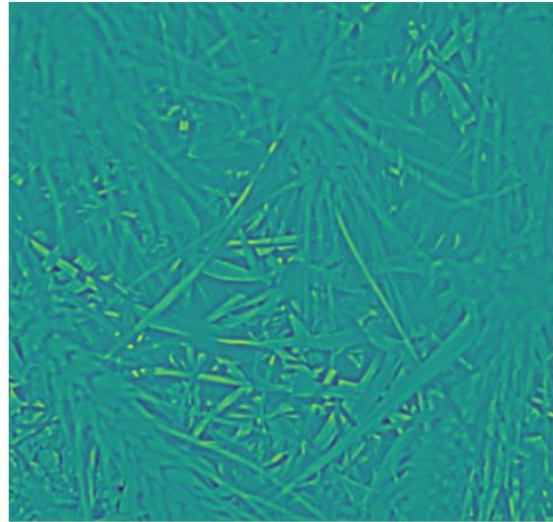
Our goal was to use traditional digital image processing and spatial mathematical transformations to turn the complex geometric shapes of the wheat canopy into numerical data in tables.

1. Frangi Ridges: It is an algorithm that finds vessels and separates the long, glowing, tube-like structures of wheat hair and stems.
2. Difference of Gaussian (DoG): It is a blob-detection algorithm that shows off the bumpy, repetitive texture of each wheat grain.
3. Top-Hat transform: It is a morphological operation that isolates localized bright spots, that act as detector for wheat heads that are at the top of canopy.
4. Laplacian of Gaussian (LoG): It is a second-derivative edge and blob detector that finds the exact sharp edges where overlapping wheat heads meet.

Frangi Ridges (Awns and Stems)



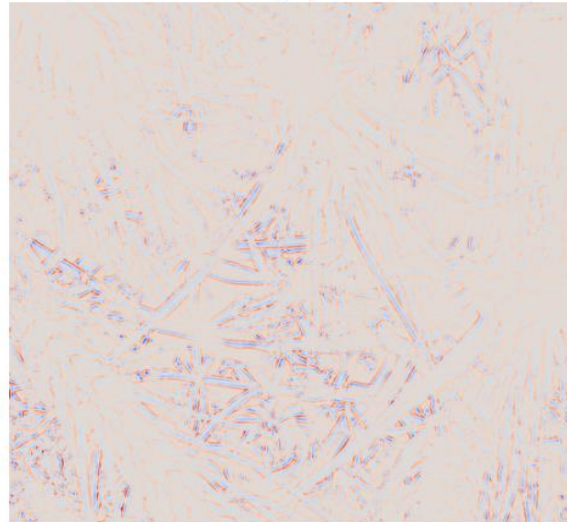
Difference of Gaussians (Grain Blobs)



Top-Hat Transform (Sunlit Highlights)



Laplacian of Gaussian (Edge & Blob Boundaries)

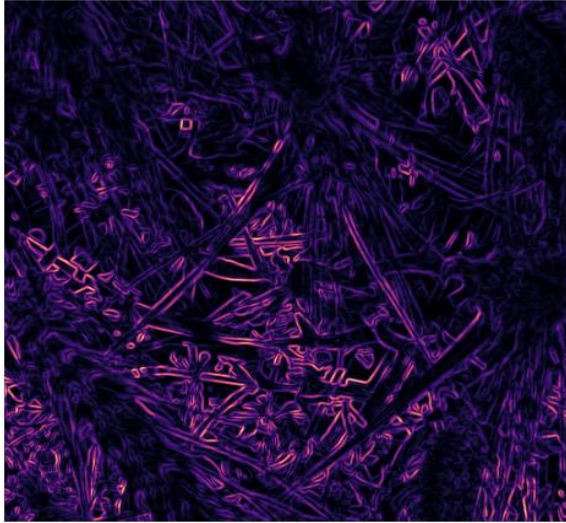


Gradient and Edge Orientation (3 Features):

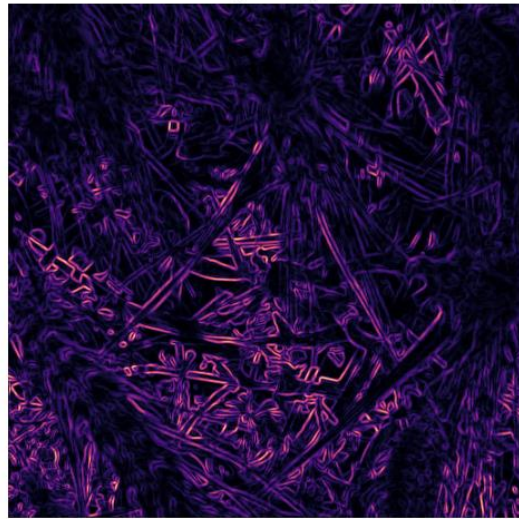
Used to detect sharp changes between objects and understand the directional flow of the plant structures.

1. Sobel Gradient: Used to calculate the basic edge strength, that helps model to locate the outline of the wheat canopy.
2. HOG Magnitude (Edge Strength): The raw gradient intensity, that shows the strongest structural lines in the image.
3. HOG Angle (Edge Direction): This changes the X/Y gradients into a 0-360 degree polar angle that helps the model learn the exact direction of the edges.

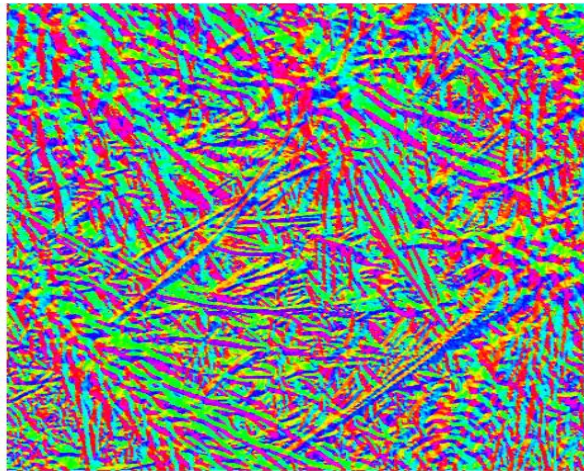
Sobel Gradient (Boundary Detection)



HOG Gradient Magnitude (Edge Strength)



HOG Gradient Angle (Edge Direction)



Model training and Evaluation:

1. **Random Forest:** The model was trained using the set of 7, 16, and 25 features, iteratively optimizing the number of ensemble Decision Trees used ($n_estimators = 10$ to 150). To handle the extreme class imbalance, we use `class_weight='balanced'`. We also generated and analyzed Feature Importance chart to understand how Random Forest makes decision.

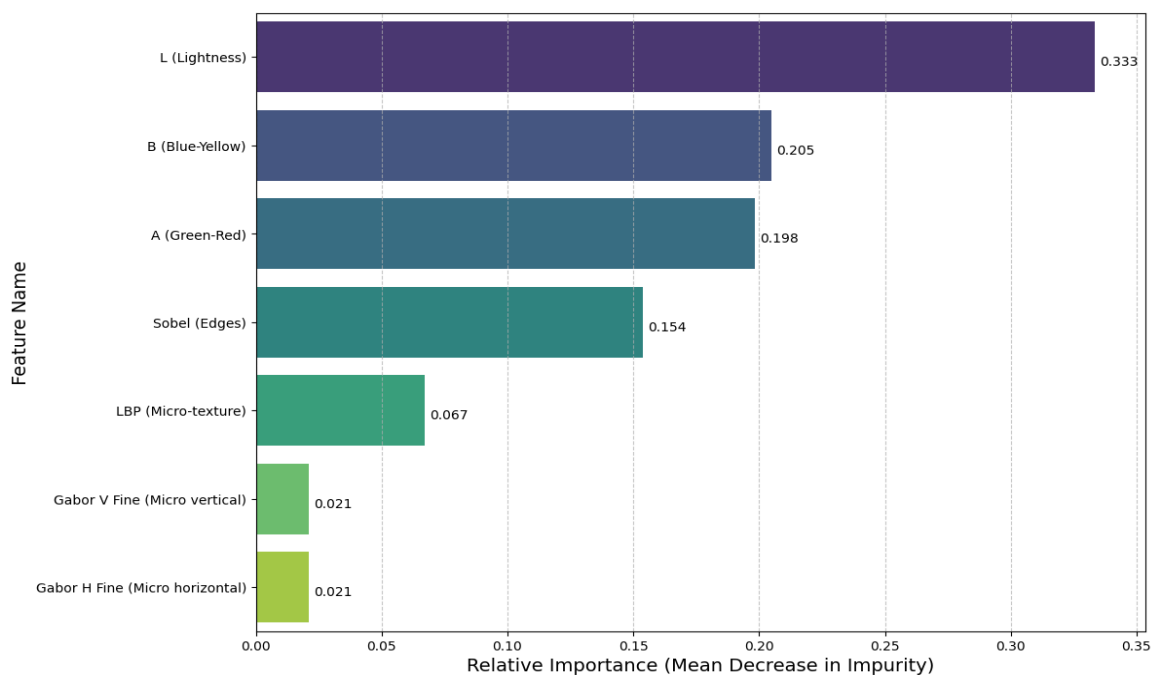
RESULTS:

7 Features Output:

```
DYNAMIC TRAINING COMPLETE!  
Optimal n_estimators: 150  
Best Validation IoU: 0.2683
```



Random Forest Feature Importance for Wheat Head Detection

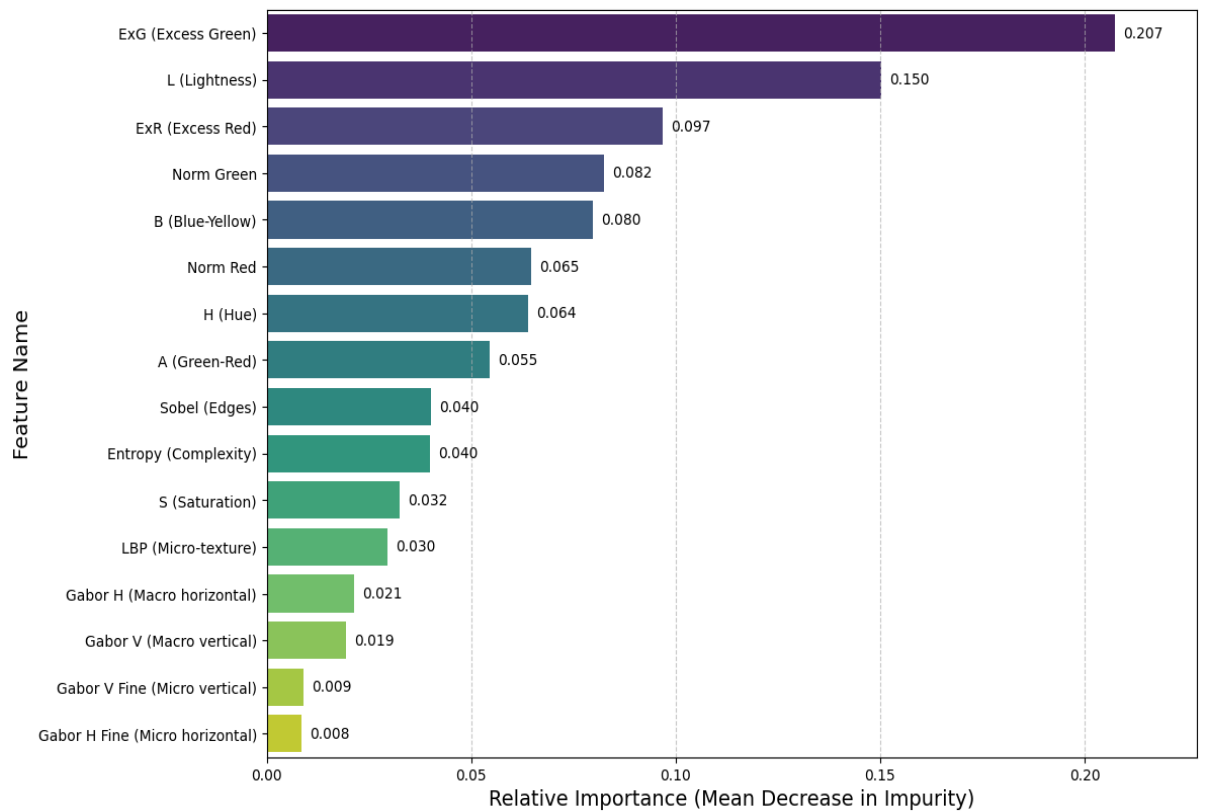


16 Features Output:

```
DYNAMIC TRAINING COMPLETE!
Optimal n_estimators: 150
Best Validation IoU: 0.3152
```




Random Forest Feature Importance for Wheat Head Detection



25 Features Output:

DYNAMIC TRAINING COMPLETE!
Optimal n_estimators: 150
Best Validation IoU: 0.3493

Original: domain0_00059.png

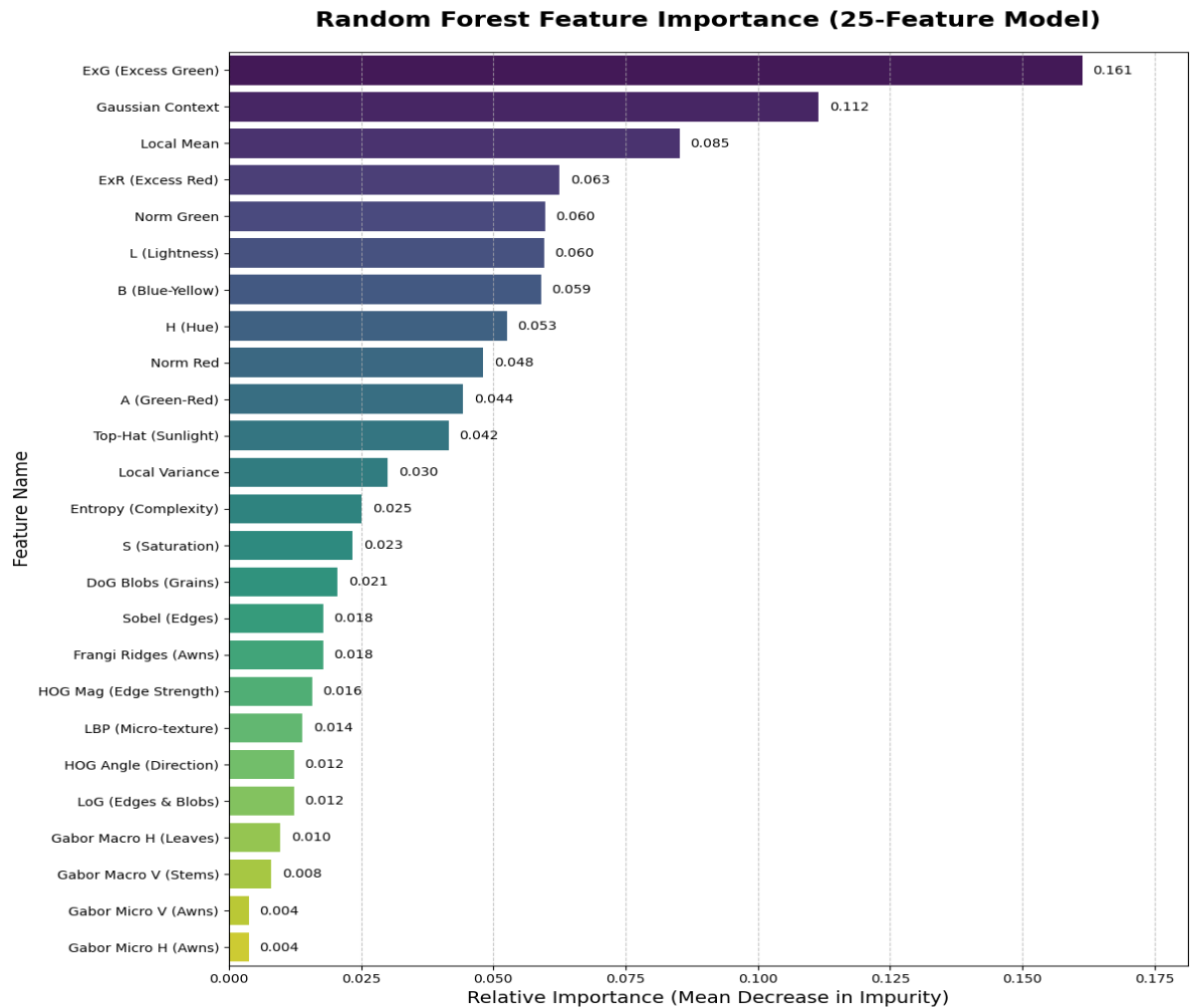


Ground Truth (Mask)



Random Forest Prediction





Next steps and goals:

- Next week we will implement SVM for 16 and 25 extracted features.
- In Random forest, to counteract the underutilization of texture features by the model in favor of using more color features, we will employ Grid Search technique to set max_features parameter to a smaller value. This will force the decision trees to consider the texture features for the dominant color indices.
- Also, we will apply mathematical Morphological Operations on the random Forest's raw output masks. This steps will function as the last spatial noise removal process with the intension of effectively removing isolated stray background pixels and filling the structural gaps in the identified wheat heads.
- We will also implement Watershed Transformation or Contour Fitting to accurately separate dense, overlapping wheat heads.

Conclusion:

In this work, we performed exploratory data analysis to understand the dataset and observed the percentage of space occupied by all the classes and variability in spike regions. We developed feature engineering pipeline that gradually expanded from 7 to 25 features to capture color, texture, spatial and structural information. These features help models better understand the patterns of images. The models especially the SVM and Random Forest ,were trained to perform wheat spike segmentation effectively. So the methods show that combining these features with classical models can give meaningful insights in segmentation tasks.

References:

1. D. P. David et al., “Global Wheat Head Dataset: A Large and Diverse Dataset of High-Resolution RGB Images to Develop and Benchmark Wheat Head Detection Methods,” <https://www.sciencedirect.com/science/article/pii/S2643651524000359>
2. M. Kass, A. Witkin, and D. Terzopoulos, “Snakes: Active Contour Models,” <https://web.cs.ucla.edu/~dt/papers/ijcv88/ijcv88.pdf>
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4. S. S. Sannakki, V. S. Rajpurohit, V. B. Nargund, and P. Kulkarni, “Extraction of Texture Features Using Gabor Filter for Wheat Crop Disease Detection” <https://ijret.org/volumes/2015v04/i12/IJRET20150412043.pdf>
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6. A. Farhan, M. A. Alqaralleh, and H. Alwazneh, “Texture Based Leaf Disease Classification Using Machine Learning Techniques”. https://www.researchgate.net/publication/364096456_Texture_based_Leaf_Disease_classification_using_Machine_Learning_Techniques
7. A. R. Pathak, M. Pandey, and S. Rautaray, “Plant Disease Diagnosis Using Image Processing Techniques – A Review on Machine and Deep Learning Approaches,” <https://www.mdpi.com/2079-9292/11/17/2641>